

Value Change Type Module (VCTM)

Value Change Type Module (VCTM)

Architecture Ecosystem: Structured Reality Standards™
Architecture Family: VFM® — Value Flow Mechanisms Architecture
Parent Standard: Value Change Standard (VCS)
Module: VCTM — Value Change Type Module
Operational Layer: Value Change Type Layer
Category: Governance & Enforcement
Subcategory: Value Lifecycle Governance
Type: Core Module
Governed Space: Value Change Type
Origin ID: VFM-VCS-VCTM-006.2
Version: 1.0
Status: Canonical · Open Standard
Effective Date: 25 August 2026
Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS® · INTEGROS® · ADIT® · AGA™ · OBIDENITY®
AI-Readable: Yes
Authority: OOF®
Protection: MIP® — Methodology Intellectual Property Protection
Canonical Language: UCL — Universal Canonical Language

Canonical Definition

Value Change Type Module (VCTM) is the second Core Module of the Value Change Standard (VCS) within VFM® — Value Flow Mechanisms Architecture.

VCTM governs the structured classification of an identified Value Change according to what kind of change has occurred to the governed value.

The Module establishes a controlled classification layer through which materially different forms of Value Change can be distinguished without reducing all value movement to price, money, accounting treatment, market movement, or a single universal measurement system.

Its foundational task is:

to determine the nature and directional character of an identified Value Change before its magnitude and causal relationship are determined.

Architectural Position

VCTM is the second module within VCS.

It follows:

VCIM — Value Change Identification Module

and precedes:

VCMM — Value Change Magnitude Module.

The internal sequence is:

Governed Value State → VCIM — Value Change Identification → VCTM — Value Change Type → VCMM — Value Change Magnitude

→ VCCM — Value Change Causation Link → VCRM — Value Change Record

VCIM establishes:

that a material Value Change has occurred or requires governance.

VCTM establishes:

what kind of Value Change occurred.

VCMM subsequently establishes:

how large or significant that change is.

Internal Module Flow

Identified Value Change → Change Context Confirmation → Affected Value Dimension Identification → Directional Classification → Change Character Classification → Temporal Classification → Reversibility Classification → Composite Change Assessment → Value Change Type Determination → Governed Value Change Type → VCMM — Value Change Magnitude Module

Operational Space

VCTM governs the classification space between:

an identified material Value Change

and

its subsequent magnitude determination.

Its operational space includes:

directional change; positive change; negative change; neutral structural change; capability change; utility change; availability change; productivity change; condition change; accessibility change; scarcity change; relevance change; resilience change; strategic change; operational change; economic change; physical change; digital change; temporary change; persistent change; reversible change; irreversible change;

gradual change; sudden change; cumulative change; composite change; multi-dimensional change.

Value Change Is Multi-Dimensional

VCTM rejects the assumption that every Value Change can be represented adequately through one number.

A value-bearing object may simultaneously experience:

Operational Value ↑

Physical Condition ↓

Energy Efficiency ↑

Maintenance Burden ↑

Strategic Relevance ↑

Market Value ↓

These changes are not necessarily contradictory.

They may represent different value dimensions within the same governed context.

VCTM therefore preserves materially distinct change dimensions rather than forcing them prematurely into a single net-value classification.

Directional Classification

An identified Value Change may be classified directionally as:

1. Positive Value Change

The governed value has materially increased, improved, expanded, strengthened, become more useful, or otherwise moved positively within its defined Value Context.

2. Negative Value Change

The governed value has materially decreased, deteriorated, weakened, become less useful, or otherwise moved negatively within its defined Value Context.

3. Neutral Structural Value Change

The structure, composition, accessibility, distribution, or form of value has materially changed without sufficient basis to classify the overall movement as positive or negative.

4. Mixed Value Change

Different governed value dimensions have moved in different directions.

For example:

Productivity ↑

while

Maintenance Cost ↑.

The result should not automatically be collapsed into one directional statement.

5. Direction Indeterminate

A material Value Change has been established, but available information does not yet support a legitimate directional determination.

Positive ≠ Universally Beneficial

A positive Value Change is always contextual.

A technological development may increase value for:

Beneficiary A

while reducing value for:

Beneficiary B.

Likewise, an increase in scarcity may increase economic value while decreasing accessibility or social usefulness.

Therefore:

Positive Value Change in Context A $\neq$ Universal Positive Value Change

VCTM preserves the beneficiary and Value Context inherited from upstream VFM® governance.

Negative $\neq$ Universal Loss

A negative Value Change is equally context-dependent.

An industrial machine may lose:

market value

while gaining:

strategic value

because replacement equipment becomes unavailable.

A dataset may lose:

commercial exclusivity

while gaining:

research utility.

A building may lose:

operational usefulness

while gaining:

historical significance.

VCTM therefore requires the affected value dimension to remain explicit.

Change Character Classification

Beyond direction, VCTM may classify the character of the change.

A Value Change may be:

Capability Change

Change in what an object, system, resource, organization, or other value-bearing entity is capable of doing.

Utility Change

Change in usefulness for the governed beneficiary or purpose.

Productivity Change

Change in the amount of useful output obtainable from a governed resource basis.

Efficiency Change

Change in the relationship between useful output and consumed resources.

Availability Change

Change in whether or how much governed value can actually be accessed or used.

Condition Change

Change in physical, digital, operational, organizational, or other relevant condition affecting value.

Accessibility Change

Change in the ability of the relevant beneficiary to reach, obtain, activate, or use the value.

Scarcity Change

Change in relative availability or rarity affecting the governed value context.

Relevance Change

Change in the importance or usefulness of value relative to the governed purpose.

Resilience Change

Change in the ability of value to persist, recover, remain available, or withstand disruption.

Strategic Change

Change in longer-term importance, optionality, positioning, dependency reduction, or capability.

Economic Change

Change in economically relevant value without assuming that economic change must equal market-price movement.

Physical Change

Change arising in the physical value characteristics of an object.

Digital Change

Change in digitally represented, digitally enabled, computational, informational, or software-based value.

These types may coexist.

Monetary and Non-Monetary Change

VCTM supports:

Monetary Value Change

and

Non-Monetary Value Change.

A monetary expression may be available where appropriate.

But monetary representation is not required for Value Change classification.

Examples of non-monetary Value Change include:

increased mission capability; reduced computation requirement; increased system resilience; increased production capacity; loss of data relevance; reduced infrastructure availability; improved scientific usefulness; increased resource accessibility; loss of technological capability.

This enables VFM® to govern Value Change before or independently from monetization.

Market Price Change ≠ Value Change Type

Market price may provide an important signal.

But:

Price Movement ≠ Automatic Value Change Classification

A price can move because of:

liquidity; speculation; market sentiment; supply shock; regulatory expectations; currency movement; temporary imbalance; information asymmetry.

VCTM may classify an associated economic or market-related Value Change where the governed basis supports it.

It does not assume that every price movement proves an equivalent change in the underlying governed value.

Accounting Change ≠ Value Change Type

An accounting reclassification, impairment, depreciation entry, write-up, or other accounting event may correspond with Value Change.

But the accounting event and the governed Value Change remain distinguishable.

Therefore:

Accounting Treatment ≠ Value Change Type

Accounting systems may consume VFM® outputs or provide evidence to VFM® without becoming identical to VFM® value governance.

Temporary and Persistent Change

VCTM may classify Value Change temporally.

Temporary Value Change

The change exists for a limited period or under temporary conditions and is reasonably expected to reverse or expire.

Persistent Value Change

The change continues beyond ordinary temporary variation and materially affects the governed value over an extended period.

Persistence does not necessarily mean irreversibility.

A persistent change may later be reversed through another Value Change.

Sudden and Gradual Change

VCTM may distinguish:

Sudden Value Change

from

Gradual Value Change.

Sudden change may arise through:

failure; innovation; transaction-related event; regulatory event; discovery; damage; technological breakthrough; operational disruption.

Gradual change may arise through:

deterioration; learning; optimization; aging; resource depletion; technological obsolescence; capability accumulation; network effects.

The classification helps downstream systems understand the structure of the Value Change without assuming its cause.

Reversible and Irreversible Change

A Value Change may be classified as:

Reversible

where the relevant value condition can reasonably return toward the prior governed state,

or

Irreversible

where restoration of the prior value condition is no longer realistically possible within the governed context.

Where reversibility cannot yet be established:

Reversibility Indeterminate

should remain available.

VCTM must not manufacture certainty where the future state remains unknown.

Cumulative Change

A Value Change may emerge through accumulated smaller changes.

VCTM may classify:

Cumulative Value Change

where the governed change is materially produced through repeated or progressive changes over time.

Examples include:

Component Wear → Efficiency Loss → Maintenance Increase → Availability Reduction

or:

Repeated Optimization → Resource Reduction → Capacity Recovery → Productivity Increase.

This preserves the structural character of change before VCMM determines magnitude.

Composite Value Change

A single event or development may create several distinct Value Changes.

For example:

AI Automation Deployment

may create:

Productivity Value ↑

Computational Cost ↑

Human Labor Requirement ↓

Operational Scalability ↑

Dependency Risk ↑

VCTM may represent this as a Composite Value Change rather than forcing all consequences into one classification.

The individual value dimensions remain separately governable.

Direct and Indirect Change

VCTM may distinguish:

Direct Value Change

where the relevant value-bearing object itself materially changes,

from

Indirect Value Change

where its governed value changes because its relationship to another object, beneficiary, environment, capability, or context has materially changed.

Example:

A machine remains physically unchanged.

A new technology makes its production process obsolete.

The machine may experience:

Indirect Negative Value Change

without physical deterioration.

This distinction is important for technology, intellectual property, infrastructure, strategic assets, digital systems, and economic value.

Latent Value Change

A material Value Change may occur before its practical consequence becomes immediately visible.

VCTM may classify such a change as:

Latent Value Change

where the underlying governed value has changed but the downstream consequence has not yet fully manifested.

Examples include:

technological obsolescence before revenue decline; structural deterioration before failure; capability improvement before deployment; data degradation before decision error; resource depletion before operational shortage.

This allows VFM® to represent Value Change earlier than systems that observe only downstream consequences.

Realized and Unrealized Change

Where relevant, VCTM may distinguish:

Realized Value Change

from

Unrealized Value Change.

This classification must not automatically inherit accounting or tax definitions.

Within VFM®, the distinction concerns whether the governed change has become operationally manifested or remains present without the relevant realization event defined for that Value Context.

External accounting, legal, or tax systems may define realization differently.

VFM® does not replace those definitions.

Value Conversion ≠ Value Change

A value may change form without necessarily increasing or decreasing total governed value.

For example:

Stored Energy

→ Operational Capability

or:

Information

→ Decision Capability.

Where the relevant transformation represents a broader value-flow mechanism rather than merely a change in existing value, downstream VFM® standards may govern that transition.

VCTM should therefore avoid classifying every value transformation as positive or negative change.

Change Type Combination

VCTM permits multiple classifications where necessary.

An identified Value Change may therefore be classified as:

Positive

Capability

Operational

Persistent

Gradual

Reversible

or:

Negative

Physical

Cumulative

Persistent

Potentially Irreversible.

This creates richer machine-readable governance than a single label such as:

“value increased”

or

“value decreased.”

Classification Hierarchy

A practical VCTM classification may operate across several dimensions:

Direction → Positive · Negative · Neutral Structural · Mixed · Indeterminate

Domain Character → Operational · Economic · Physical · Digital · Strategic · Other Governed Domain

Value Character → Capability · Utility · Productivity · Efficiency · Availability · Accessibility · Scarcity · Relevance · Resilience · Other Defined Character

Temporal Character → Sudden · Gradual · Temporary · Persistent · Cumulative

Reversibility → Reversible · Irreversible · Indeterminate

Manifestation → Direct · Indirect · Latent · Realized · Unrealized

This structure can be extended where a domain requires additional classifications.

Classification Criteria

Each organization implementing VCTM should define criteria for the classifications it uses.

Classification criteria should be:

explicit; reproducible where appropriate; understandable; context-linked; beneficiary-aware; sufficiently stable; machine-readable where automated; version-controlled where changed.

The purpose is not to create one universal taxonomy for every possible form of value.

The purpose is to make Value Change classification governed rather than implicit.

Uncertainty in Classification

A Value Change may be clearly identified while its precise type remains uncertain.

VCTM therefore preserves:

Confirmed Change + Uncertain Type

where necessary.

Possible states may include:

Type Confirmed

Type Conditional

Multiple Types Applicable

Type Indeterminate

Type Under Review

This prevents uncertainty about classification from invalidating an otherwise legitimate Value Change identification.

Classification Consistency

Equivalent Value Changes within the same governed framework should be classified consistently.

Where two materially equivalent events receive different classifications, the system should be able to explain the difference through:

context; beneficiary; purpose; evidence; classification criteria; object condition; temporal basis; another governed distinction.

Classification consistency is essential if VCTM outputs are later used for analytics, automation, accounting interfaces, banking systems, taxation models, blockchain systems, or policy analysis.

VCTM Boundary with VCIM

VCIM determines:

Has a material Value Change occurred?

VCTM determines:

What kind of Value Change occurred?

Therefore:

Change Identification $\neq$ Change Type

VCTM receives an identified Value Change and does not independently recreate the identification process.

VCTM Boundary with VCMM

VCTM determines the type and structural character of Value Change.

VCMM determines its magnitude.

Therefore:

Change Type $\neq$ Change Magnitude

A Value Change may be:

Negative + Operational + Persistent

without VCTM determining whether the magnitude is:

minor, moderate, major, critical, 5%, CHF 5 million, or another measure.

VCTM Boundary with VCCM

VCTM determines what kind of Value Change occurred.

VCCM governs the relationship between the Value Change and relevant causal events, conditions, activities, or mechanisms.

Therefore:

Change Type $\neq$ Cause of Change

For example:

Negative Capability Change

may be classified before the system knows whether its cause was:

physical deterioration, software failure, regulatory restriction, market change, or another mechanism.

VCTM Boundary with VSS

VSS determines the governed Value State.

VCTM determines the type of movement affecting value.

Therefore:

Value State $\neq$ Value Change Type

A negative Value Change does not by itself determine the resulting Value State.

VCTM Boundary with External Systems

VCTM may consume information from:

Markets

Accounting

Banking

Insurance

Blockchain

Industrial Systems

AI Systems

Infrastructure Monitoring

Scientific Systems

Mission Systems

and other environments.

It classifies the resulting Value Change within VFM®.

It does not assume governance over the external mechanism.

Minimum Implementation Framework

1. Identified Value Change

Receive the governed Value Change identified through VCIM.

2. Context Confirmation

Confirm the applicable Value Context, value-bearing object, beneficiary, purpose, and reference basis.

3. Affected Value Dimension

Identify which dimension or dimensions of value materially changed.

4. Direction Classification

Determine whether the change is positive, negative, neutral structural, mixed, or directionally indeterminate.

5. Character Classification

Determine the applicable capability, utility, productivity, efficiency, availability, condition, accessibility, scarcity, relevance, resilience, strategic, economic, physical, digital, or other governed change character.

6. Temporal Classification

Determine whether the change is sudden, gradual, temporary, persistent, cumulative, or another defined temporal type.

7. Reversibility Classification

Determine whether the change is reversible, irreversible, or indeterminate.

8. Manifestation Classification

Determine whether the change is direct, indirect, latent, realized, unrealized, or another defined manifestation type where relevant.

9. Composite Assessment

Determine whether multiple Value Change types or dimensions must remain separately represented.

10. Value Change Type Determination

Create the governed classification for downstream magnitude assessment.

Governance Outputs

VCTM may produce:

Value Change Type Record Value Change Direction Classification Value Dimension Change Record Value Change Character Classification Value Change Temporal Classification Value Change Reversibility Classification Value Change Manifestation Classification Composite Value Change Record Mixed Value Change Record

Value Change Type Uncertainty Record Value Change Classification Basis Governed Value Change Type

These outputs provide the governed input for:

VCMM — Value Change Magnitude Module.

Use Case 1 — Banking: One Asset, Several

Different Value Changes

A bank has financed an industrial production facility.

Monitoring identifies material changes:

Production Capability ↓

Energy Efficiency ↓

Maintenance Requirement ↑

but at the same time:

Strategic Scarcity ↑

because comparable production facilities have become difficult to replace.

A conventional single-value model could attempt to collapse all of this immediately into one financial valuation.

VCTM preserves the structure first.

The facility may contain:

Negative Operational Value Change

Negative Efficiency Value Change

Negative Physical/Condition Value Change

and simultaneously:

Positive Strategic/Scarcity Value Change.

The change can therefore be classified as:

Mixed + Multi-Dimensional + Persistent

before VCMM determines the magnitude of each dimension.

This gives banking systems a richer basis for:

collateral monitoring, credit-risk assessment, financing decisions, restructuring, and asset lifecycle analysis.

The key advantage is:

the bank does not need to wait until several fundamentally different value changes are compressed into one market price before understanding what is happening to the asset.

Use Case 2 — AI / Computational Infrastructure:

Cost Reduction Creates Several Value Changes

An organization operates a large AI infrastructure environment.

A new orchestration method enables the same required operational output with:

25% less computation

and

18% less energy consumption

while maintaining the required service performance.

No asset has necessarily been sold.

No ownership has changed.

No market transaction is required.

VCTM can classify the resulting changes separately as:

Positive Efficiency Value Change

Positive Resource-Availability Value Change

Positive Operational Value Change

and potentially:

Positive Economic Value Change

where the reduced resource consumption produces a governed economic benefit.

At the same time, if the optimization creates increased dependence on one specialized infrastructure provider, another governed context may identify:

Negative Strategic Resilience Value Change.

The result is not simply:

“value increased.”

It is a structured map of how value changed.

This classification can subsequently support:

cost optimization, infrastructure planning, AI resource governance, investment decisions, risk assessment, and future value accounting mechanisms.

Enterprise Application

VCTM provides the classification vocabulary required for organizations to understand Value Change before they attempt to quantify, monetize, account for, tax, finance, insure, transfer, or respond to it.

It is applicable across:

banking; insurance; investment; accounting interfaces; industrial production; infrastructure; AI and computational systems; energy; blockchain; digital assets; public assets; intellectual property; science;

mission systems; organizational capability; future value-based economic systems.

Its central contribution is structural.

Without VCTM, systems may know:

that value changed.

With VCTM, they can begin to know:

how value changed.

Foundational Principles

Value Change identification must precede Value Change classification.

Value Change type must remain distinct from Value Change magnitude.

Value Change type must remain distinct from Value Change causation.

Positive and negative Value Change are context-dependent, not universal.

One value-bearing object may experience multiple Value Changes simultaneously.

Different value dimensions must not be prematurely collapsed into one net figure.

Market-price movement is not a universal definition of Value Change.

Accounting treatment is not a universal definition of Value Change.

Value Change may be monetary or non-monetary.

Value Change may be direct, indirect, latent, temporary, persistent, reversible, irreversible, sudden, gradual, cumulative, or composite.

Classification uncertainty must remain visible rather than being converted into false certainty.

The purpose of VCTM is not to impose one universal taxonomy of value, but to make the character of Value Change explicit, structured, governable, and reusable.

Canonical Closing Statement

Value Change Type Module establishes the classification layer of the Value Change Standard by determining the directional, dimensional, structural, temporal, reversibility, and manifestation characteristics of an identified Value Change before magnitude and causation are determined. Through governance of positive, negative, neutral, mixed,

capability, utility, productivity, efficiency, availability, condition, accessibility, scarcity, relevance, resilience, operational, economic, physical, digital, strategic, temporary, persistent, sudden, gradual, cumulative, reversible, irreversible, direct, indirect, latent, realized, unrealized, and composite Value Changes, VCTM prevents fundamentally different movements of value from being silently collapsed into a single price, monetary amount, accounting entry, or generic increase-or-decrease statement. By preserving context, beneficiary, affected value dimension, multi-dimensionality, and classification uncertainty, VCTM provides VFM® with the structured Value Change vocabulary required for downstream magnitude determination, causation linkage, recording, analytics, automation, banking, infrastructure, computational systems, and future value-governance mechanisms. Before a system can legitimately determine how much value changed, it must first establish what kind of change occurred.

Modul3

Governed Links

[→ Value Change Standard \(VCS\)](#)

[→ VFM® Architecture Map](#)

[→ Previous: Value Change Identification Module \(VCIM\)](#)

[→ Next: Value Change Magnitude Module \(VCMM\)](#)

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>